

HCMTT: Hybrid Clustering for Multi-Target Tracking in Wireless Sensor Networks

Faezeh Hajiaghajani, Marjan Naderan, Hossein Pedram, Mehdi Dehghan

Computer Engineering and Information Technology Department,
Amirkabir University of Technology (Tehran Polytechnic)
Tehran, Iran

{haghajanim, naderan, pedram, dehghan}@aut.ac.ir

Abstract— Mobile Target Tracking is viewed as one of the primary applications in Wireless Sensor Networks (WSNs). Due to WSNs' special characteristics, there happens to be a trade-off between tracking accuracy and energy consumption. We propose an efficient multi-target tracking method in which the tracking task is switched between static clusters as the backbone of the network and dynamic clusters, which form in boundary regions. To reduce energy dissipation in case of extra dynamic cluster formations, we apply a merging mechanism where intersecting dynamic clusters appear. The numerical results achieved using the Castalia simulator verify the efficiency of the proposed scheme comparing to HCTT algorithm, with 10% decrease in the average power consumption and an average of 36.4% decrease in the number of redundant data packet transmissions at the cost of 1.5% increase in the control packets' overhead. Satisfying results are also obtained for tracking quality of our method compared to a prediction-based method, DPT.

Keywords- Object tracking sensor networks; dynamic clustering; merging.

I. INTRODUCTION

Mobile Object Tracking has gained much attention during recent years as the basis of many important applications in Wireless Sensor Networks (WSNs) such as field surveillance, intruder tracking, and habitat monitoring [1,2]. It mainly refers to obtaining a record of the trajectory of moving objects over space and time by processing sensory data. Generally, WSNs' design and resource constraints such as short communication range, limited processing power, restricted storage in nodes, low bandwidth, and limited energy resources make design and implementation of tracking algorithms difficult. In particular for multi-target tracking applications, large communication overhead resulting from packet exchanges among neighbour nodes and multi-hop communications turn the task of mobile object tracking critical in terms of energy consumption [3]. Regarding the network structure point of view, object tracking protocols in WSNs can be classified into three main categories: leader-based protocols, tree-based protocols and cluster-based protocols [4]. Two other groups, i.e., Mobicast message-based and hybrid methods are added to this classification in [5]. Each protocol in the aforementioned categories may be combined with prediction strategies, which reduce the energy consumption of nodes

[5]. Since only particular nodes close to the object trajectory are active in prediction strategies, prediction errors can lead to a higher target missing ratio as a result of awakening the wrong nodes. This fact highlights the trade-off between energy conservation and tracking accuracy as the most important challenge in prediction-based tracking strategies [6]. In other words, as the number of active sensors increases, an object would less likely get lost due to sufficient sensing reports. On the other hand, a greater number of active nodes results in supplementary energy dissipation. Most of the existing target tracking methods in WSNs are basically provided for single target tracking (e.g. [7, 8]). However, multi-target tracking seems to be of a great importance due to the variety of its real applications, for instance, a large natural habitat in a downtown area of a city (e.g. [9]) or wildlife tracking (e.g. [10]). In case of multi-target tracking algorithms, energy conservation becomes more challenging as more numbers of nodes take the responsibility of tracking in the network. One interesting idea to reduce the number of active nodes in cluster-based multi-target tracking algorithms is to merge redundant overlapping clusters [11].

In this paper, we propose a distributed scalable target tracking algorithm which tracks multiple targets more efficiently comparing to hybrid clustering methods such as HCTT [7] and more accurately compared to prediction based strategies such as DPT [16]. We use static clusters to address scalability goals, as well as dynamic clusters to take advantage of sensors collaboration and information sharing especially in boundary regions of static clusters. Despite hybrid clustering increases the overall overhead of the algorithm, it removes objects' misses (observed in the case of prediction errors) and increases tracking accuracy. We propose a mechanism to merge and separate dynamic clusters, formed around objects which move in a same region. Comparison of our tracking algorithm with HCTT, which does not consider the issue of cluster merging and multi-target tracking, reveals 10-15% savings in energy consumption.

The rest of the paper is organized as follows: we give an overview of a number of existing protocols in section II. We introduce and define existing problems in section III. We present our new method and delve into its details in section IV and discuss the simulation results in section V. Finally

we conclude the paper and give the future work in section VI.

II. RELATED WORK

In this section, we give a brief discussion on classification of schemes according to [5], with examples including some recent methods. In prediction-based schemes, movement of an object can be predicted as long as it traverses within a particular speed range. PES [12], anticipates the future movement of an object so that only the sensor nodes, expected to detect the object, will be activated leading to less energy dissipation. Xu et al. proposed a dual prediction-based reporting mechanism in [13], in which both sensor nodes and base station predict the future location of moving object. Sensors avoid extra reading transmissions, unless the predicted position of the object differs from its real one. Apparently, all prediction methods are prone to estimation error which might lead to activation of wrong sensors. Besides, recovery of the lost object requires extra energy consumption.

Among the tree-based schemes, DCTC [8] relies on a convoy tree structure which is formed and pruned dynamically while tracing the object's trajectory. Optimizing the tree coverage and finding the most efficient convoy tree in DCTC are challenging problems. In DOT [14], nodes exchange spatial neighbours' location in a Gabriel graph to recognize the closest sensor to the moving object. Yen et al. proposes analytic estimates of mobility profiles based on Markov-chain model in [15] and indicates that the problem of finding an optimal message report tree that requires the least amount of report messages is NP-hard.

In cluster-based methods, cluster heads are responsible for gathering sensing reports from their members as well as localizing the target. Static clustering takes place statically at the deployment phase of network and according to some measures such as nodes' geographic locations. Since network's structure does not change in static clustering methods, they do not take the targets' dynamics over time. Instead, in dynamic clustering methods, dynamic clusters emerge surrounding moving target. Rare [24] is an example of a static clustering target tracking method which addresses two issues concerning the data, i.e., low quality data which occurs when sensors are far from the target and redundant data due to overlapping sensing areas. Yang et al. proposed a scalable prediction-based protocol, DPT, which merely applies dynamic clusters. At any given time, the optimal set of active sensors are determined by a prediction method, to minimize the overall power consumption.

Hybrid clustering methods such as CODA [18] and HCTT exchange the task of tracking between dynamic and static clusters. In CODA, moving boundaries of continuous objects are detected and tracked throughout the network. As the object increases in size and the boundaries move, new dynamic clusters appear to track the object. In HCTT, a single target tracking method, boundary nodes of static

clusters contribute to dynamic cluster formations, when the object reaches edges of static clusters. HCTT outperforms prediction-based methods in case of robustness because it does not rely on prediction schemes to trigger dynamic clusters and is not therefore prone to prediction errors. However, it consumes more energy proportionally, due to extra dynamic cluster formations.

III. PROBLEM DESCRIPTION

Several important factors influence the design of a target tracking algorithm, some of which are specified in [4]. To design our method, we consider the following issues mainly:

A. Robustness

Target loss and providing a recovery mechanism are crucial issues in analyzing target tracking algorithms. However, recovery mechanisms are not well defined in many of the proposed schemes such as [19]. DPT claims to be robust against prediction failures by applying a hierarchical recovery mechanism in which active sensors, switch to high beam to find the missing target. If they do not succeed in detection, additional sensors are activated to help. PES, analogously to DPT, awakens particular nodes around the predicted trajectory of the missing object. If unsuccessful, it floods a network-wide awakening message to activate all sensors. Evidently, all similar recovery mechanisms suffer huge energy consumption resulting from an increase in the number of active nodes.

A preventive solution aims to inhibit target loss in risky locations. HCTT emphasizes boundary problem, which points to a lack of collaboration among sensor nodes along boundaries of static clusters. As a solution, this method chooses on-demand dynamic clustering near edges of static clusters. To preserve the tracking quality, we utilize dynamic clusters in boundary regions, as well.

B. Power Conservation

1) *Zigzag Movement*: Zigzag movement of the target in boundary regions, causes frequent formation and dismissal of redundant dynamic clusters, which leads to extra transmissions of control messages and energy dissipation. In HCTT, zigzag movement of an object between a static cluster and a dynamic one, is a relevant example.

2) *Overlapping Dynamic Clusters*: Overlapping dynamic clusters in some tracking schemes such as [17] and our proposed method, result in extra data transmissions from nodes to sink, hence, considerable energy consumption. Therefore, dynamic clusters should be carefully formed in the right place and at the appropriate time. To prevent redundant message transmissions in [17], a new dynamic cluster head election algorithm is presented. Instead, we apply a merging technique in order to reduce redundant dynamic clusters.

IV. PROPOSED ALGORITHM

We assume that the network is initially clustered into static clusters applying a simple clustering method as follows: static cluster heads (SCH) are selected based on their geographic locations. Sensor nodes choose the nearest SCH, based on the signal strength they receive through a join message (JOIN_SC). Each node replies to the first join message it receives and sets its permanent static cluster ID. Finally, SCHs form member tables by receiving join confirmations. Inspired by the definition of boundary regions in HCTT, we define “boundary nodes” as those members of static clusters, which have neighbours from a different cluster. The rest of the nodes that form the internal section of static clusters are named “internal nodes”. As the objects move within the internal section of a static cluster, internal nodes transmit sensing reports to their SCHs. When the object approaches boundaries of the static cluster, a dynamic cluster is formed on demand to continue tracking in boundary regions, where target loss probability is the highest. We also assume that association is done in cluster heads, hence that is not an issue to discuss within the scope of this paper. Below we describe formation and dismissal of dynamic clusters.

A. Dynamic Cluster Formation

Upon receiving the first sensing report from a boundary node, a static cluster head discovers that the object is moving towards boundaries and might leave the static cluster. It then sends a U_R_DCH message to the closest boundary node to the object and elects it as a dynamic cluster head (DCH). The designated DCH subsequently forms a dynamic cluster by broadcasting a JOIN_DC message to its neighbours, generally boundary nodes of two or more static clusters. The JOIN_DC message works as an activation announcement for sleeping members. One-hop neighbours, which receive this message, reply back to the DCH (JOIN_CONFIRM). The object then follows one of the following behaviours: 1) it changes direction and moves back into former static cluster or 2) it moves further into the emerged dynamic cluster. In case of the first movement model, the dynamic cluster is no longer qualified to track the target. Thus, it sends a RESIGN message to its members and collapses. Meanwhile, the static cluster continues tracking the target. In case of the second movement model, the elected dynamic cluster head prepares to start tracking the target, once it receives a sensing report from one of its members. Subsequently, the DCH sends a WORK message to its members as well as a hand off request message, HANDOFF_REQ, to the former static cluster head. Upon receiving the HANDOFF_REQ, the static cluster head assigns the task of tracking to the newly activated dynamic cluster (HANDOFF_CONF). Afterwards, those members of the static cluster, which do not belong to any dynamic cluster, are resigned and sleep.

B. Dynamic Cluster Dismissal

Dynamic cluster dismissal takes place whenever the dynamic cluster quits tracking the object. When target moves out of a dynamic cluster, member nodes receive a signal strength lower than some accepted threshold, S , and send a WAKE_UP message to their neighbours. If the object moves along the boundaries, the alarmed nodes form a new dynamic cluster. Contrarily, if the object moves into a static cluster, activated neighbours, which are mostly internal nodes of the static cluster, send sensing reports to their SCH. In either case, if the old DCH does not receive any sensing report within a time interval, T , it sends RESIGN messages to its members and collapses. Since fewer nodes are activated in this dismissal method, less energy is consumed compared to the dismissal strategy in HCTT, where the whole static cluster starts working. (Fig. 1)

C. Merging Dynamic Clusters

To save energy and reduce redundant message exchanges in overlapping clusters, we apply a merging mechanism. When two objects move close to each other such that their dynamic clusters overlap, sensors transmit sensing reports to both DCHs (Fig. 2(a)). After the merging process, members of the merged cluster send sensing reports to the selected DCH (Fig. 2(b)). The merging process begins when at least one sensor node belonging to more than one dynamic cluster senses the object(s). It instantly introduces itself as a potential cluster head, to its dynamic cluster heads by a MERGE_REQ message. The DCHs wait for some specified time, T_{MREQ} , to receive all probable MERGE_REQ messages from intersected members. Next, they choose the closest member to the target(s) as the new DCH (MERGE_CONF). The elected node, forms a new dynamic cluster; it also sends a DISMISS message to each DCH, from which it had received confirmation message. Afterwards, the former DCHs send RESIGN messages to their members and collapse. Table I, shows a summary of the control messages used in the merging process.

D. Splitting the Merged Clusters

As objects step far from each other and reach borders of the merged cluster, they would probably leave it. As mentioned in section IV-B, neighbours of dynamic cluster members, wake up and track the object, having the role of members of a new dynamic cluster or a static one.

E. Analytical Model

To discuss energy consumption in our merging method, we propose an analytical model of energy usage based on [23]. Our model differs from [23] in the sense that we evaluate the energy consumption in dynamic clusters instead of static ones. Using this energy model, we assume that the energy used by nodes to receive (E_{rx}) is a function of the number of bits received (k). The consumed energy to transmit for each node (E_{tx}) is a function of k and the square of the transmission distance.

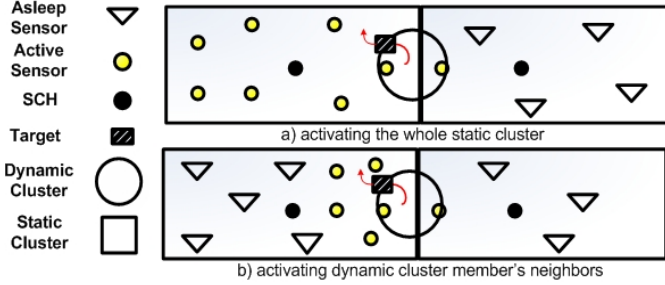


Fig. 1. Active nodes in two different methods of dynamic cluster dismissal, a) HCTT method, b) our modification.

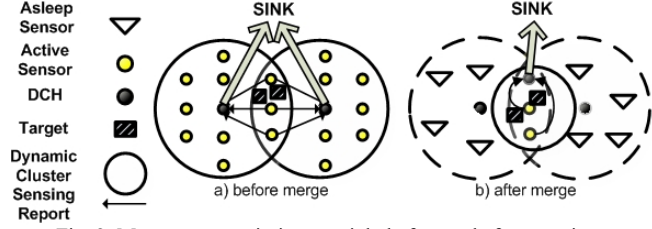


Fig. 2. Message transmissions to sink, before and after merging.

The constants E_{elec} and ϵ_{amp} depend on the physical characteristics of the transceiver hardware. If $E[M]$ is the expected number of dynamic cluster members, a dynamic cluster head receives $E[M]$ sensing reports per sample and relays them to the base station. Let d_{ABS} be the distance to the base station, averaged across all nodes and r_{ch} be the distance between a dynamic cluster member and its head. Since dynamic clusters are one hop clusters, r_{ch} is always less than the transmission range of a sensor. We also assume that each cluster head expends a constant amount of energy (E_{chm}) for each member unrelated to receiving and a fixed amount of energy per each sensing report (E_f) independent of the number of members. Thus, the energy usage per each sensing report request, i.e., sample, for a dynamic cluster head, is

$$E_{ch} = E_{tx}(k, d_{ABS}) + E[M](E_{rx}(k) + E_{chm}) + E_f \quad (1)$$

Since members only transmit reports to their DCH(s), we have the following expression for each node's energy usage, E_m .

$$E_m = E_{tx}(k, r_{ch}) + E_f \quad (2)$$

We assume that member nodes are evenly distributed among dynamic clusters and that dynamic cluster heads are selected with a p probability. Hence, the expected number of members per dynamic cluster is shown in equation 3.

$$E[M] = \frac{(1-p)n}{p} = \frac{1}{p} - 1 \quad (3)$$

Putting the results together, we get the final expression, E , which indicates the energy consumption in a dynamic cluster for each sample transmission to the base station.

$$E = p[k(\epsilon_{amp}d_{BS}^2 + E_{elec}) + (1/p - 1)(kE_{elec} + E_{chm})] + E_f + (1-p)(1/p - 1)[k(\epsilon_{amp}r_{ch}^2 + E_{elec}) + E_f] \quad (4)$$

Obviously, E is multiplied by the number of working dynamic clusters in methods such as HCTT, where separate

TABLE I. A SUMMARY OF CONTROL MESSAGES

Message Name	Description
MERGE_REQ	An intersected node informs its DCHs that it can form a merged cluster.
MERGE_CONF	Old DCHs choose the closest member node to the object as a new DCH for the merged cluster.
JOIN_DC	Elected DCH requests its neighbours to join dynamic cluster.
DISMISS	DCH of a merged cluster asks old overlapping dynamic clusters to collapse.
RESIGN	DCH asks its members to resign before collapsing at the dismissal phase.

clusters are formed per object. This proves less energy consumption in presence of fewer dynamic clusters in our merging-enabled scheme.

V. SIMULATION RESULTS

We have used the Castalia framework [20], which is based on the OMNeT++ event-based simulator [21], to evaluate the performance of our proposed method. Radio layer behavior follows [22] and a simple CSMA/CA MAC protocol is implemented in the MAC layer. To take advantage of the merging technique, we assume that the network is densely populated. Although our method has no restrictions on employing random placement of nodes, we used a grid structure to have full coverage of the area. Radio range of each sensor node is 10m and each experiment runs for 3600 seconds. Our mobility model is the Random Waypoint. We take data samples from sensors at twelve different time slots. We compare our proposed method, HCMTT, to HCTT algorithm, in which no merging takes place. We also implement DPT as a prediction based scheme, to evaluate the tracking quality of our method.

A. Power Consumption

Fig. 3 shows the average power consumption when eight objects, which lead to eight intersecting clusters exist in HCMTT. A dynamic cluster per target is formed in HCTT, as it does not provide any specific method for multi-target tracking. Results indicate that the fraction of average power consumption is always lower in the proposed method, since fewer nodes are active for tracking the target. In other words, energy consumption depends on how long the radio of each node is on, either receiving or sending packets or being idle [20]. When two dynamic clusters join into one cluster, a larger section of neighbouring static cluster members enter the sleeping phase. Moreover, as the number of nodes increases, the consumed energy is divided by a greater number, which results in the decreasing trend of power consumption. Fig. 4 illustrates the average power consumption for 400-4000 nodes, with varying number of targets. As can be seen, HCMTT saves more energy due to the merging process which uses a power saving mechanism by activating minimum number of nodes. The figure also

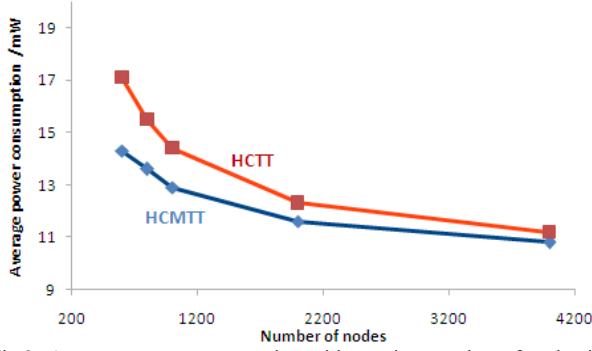


Fig.3. Average power consumption with varying number of nodes in the presence of 8 objects.

shows the outstanding behavior of DPT, in terms of power efficiency which is due to the fact that DPT does not apply static clusters. Note that, the power consumption for DPT is ideally measured, assuming that a cluster head chooses the sensor-triplet with a low beam. Thus, energy dissipation caused by high beam searching for the sensor-triplet and recovery mechanism is not considered.

B. Tracking Quality

As depicted in Fig. 4, DPT expends much less power compared to hybrid clustering methods, HCTT and HCMTT. Considering the trade-off between the tracking quality and the energy consumption in OTSNs, here we compare our algorithm with DPT according to two indicators of tracking quality. Fig. 5 illustrates the normalized mean distance between the real location of the target and the activated sensor, considering different prediction error values for DPT. It can be seen that HCMTT does not show drastic changes since it does not rely on a prediction strategy. The values of the horizontal and vertical axes are normalized by x and $2x$, respectively, in which x is the maximum distance a target moves further from its previous location with speed S in a time slot T (we set $S=1$ m/s and $T=10$ s). We also measure the mean squared error (MSE) of the distance between the activated sensor at the estimated location of the target and the most appropriate sensor, which is the closest node to the real location of the target. As depicted in Fig. 6, MSE increases in DPT as the prediction error takes greater values. HCMTT has a MSE value of zero since the closest sensor to the target is always selected as a dynamic cluster head or a reporting sensor.

C. Packet Exchanges

Packet exchanges include both data packets (sensing reports sent to cluster heads) and control packets. Fig. 7 illustrates a greater number of data packet transmissions in HCTT comparing to our method. On average, 36.4% decrease is observed in the number of sensing reports circulating in the network. As two dynamic clusters merge into one cluster, the number of remitted sensing reports reduces by half approximately.

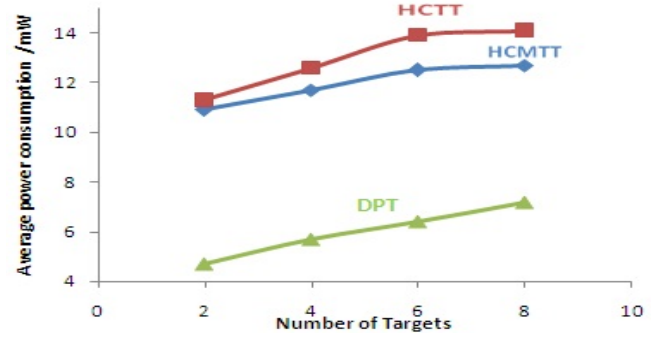


Fig.4. Average power consumption with varying number of targets.

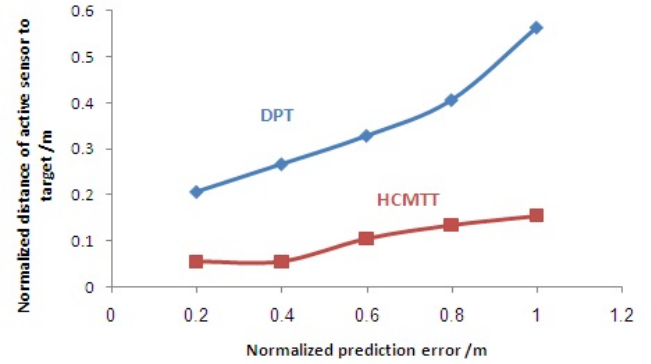


Fig.5. Proximity of active sensor to target with varying prediction error values.

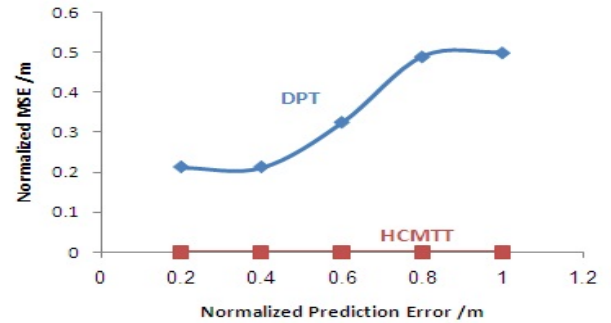


Fig.6. MSE of sensor selection with varying prediction error values.

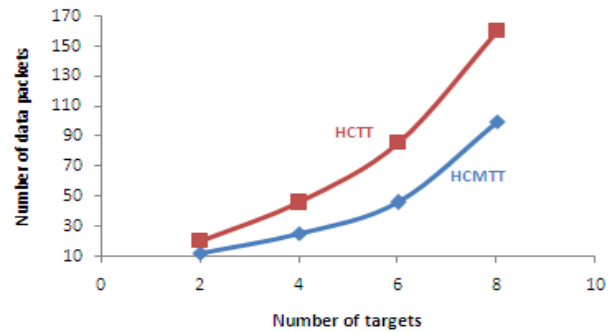


Fig.7. Total number of data packets transmitted to cluster heads when the number of targets varies.

This result is also acknowledged by the analytical model proposed in section IV.E. Furthermore, as the number of targets increases, more nodes participate in broadcasting data packets, hence, the difference between our method and HCTT becomes noticeable. From the viewpoint of control messages, it is obvious that our proposed merging process involves extra transmissions of control packets as mentioned in section IV-C. In figure 8, we illustrate the number of additional control packets in our proposed method and HCTT as the number of targets increases. The figure acknowledges slight increase of 1.5% for control packets overhead in the proposed method due to the merging cost.

I. CONCLUSION AND FUTURE WORK

In this paper, we proposed a multi-target hybrid clustering method for target tracking in WSNs. By forming dynamic clusters in boundary regions of static clusters, we reduce the risk of target miss in border lines. This technique has proved to be helpful in similar approaches, as it prevents target loss due to possible prediction errors. To bring down extra data packet transmissions to the sink node, we apply an efficient merging method for intersecting dynamic clusters. This technique enhances the overall energy consumption as it allows more sensor nodes to save energy during sleep mode. Simulation results show that our method achieves an improved behavior in terms of energy consumption comparing to a similar approach, HCTT. For future work, we plan to reduce the cost of dynamic cluster formations, when the target moves back and forth between a static cluster and a dynamic one.

REFERENCES

- [1] A. Cerpa, J. Elson, D. Estrin, L. Girod, M. Hamilton, and J. Zhao, "Habitat monitoring: Application driver for wireless communications technology," in *Proc. of the ACM SIGCOMM Workshop on Data Communications in Latin America and the Caribbean*, April 2001, pp. 20-41.
- [2] A. Mainwaring, J. Polastre, R. Szewczyk, D. Culler, and J. Anderson, "Wireless sensor networks for habitat monitoring," in *ACM International Workshop on Wireless Sensor Networks and Applications (WSNA'02)*, Atlanta, GA, September 2002, pp. 88-97.
- [3] Y.-C., Tseng, S.-P., Kuo, H.-W., Lee, C.-F., Huang, "Location Tracking in a Wireless Sensor Network by Mobile Agents and Its Data Fusion Strategies," *The Computer Journal*, Vol. 47, NO. 4, pp. 448-460, 2004.
- [4] Naderan, M., Dehghan, M., Pedram, H. and Hakami, V. (xxxx) 'Survey of mobile object tracking protocols in Wireless Sensor Networks: a network-centric perspective', *Int. J. Ad Hoc and Ubiquitous Computing*, Vol. x, No. x, pp.xxx-xxx.
- [5] S. Bhatti, J. Xu, "Survey of target tracking protocols using wireless sensor network," in *Proc. of ICWMC*, 2009, pp. 110-115.
- [6] W.-C., Yang, Z. Fu, J.-H., Kim, M.-S., Park, "An adaptive dynamic cluster-based protocol for target tracking in wireless sensor networks," in *Proc. of the APWeb/WAIM'07*, 2007, pp. 156-167.
- [7] Z. Wang, W. Lou, Z. Wang, J. Ma, and H. Chen, "A novel mobility management scheme for target tracking in cluster-based sensor networks," in *IEEE DCOSS*, Jun. 2010, pp. 172-186.
- [8] W. Zhang, G. Cao, "DCTC: Dynamic Convoy Tree-based Collaboration for Target Tracking in Sensor Networks," *IEEE Trans. on Wireless Comm.*, Vol. 3, NO 5, pp. 1689-1701, Sept., 2004.

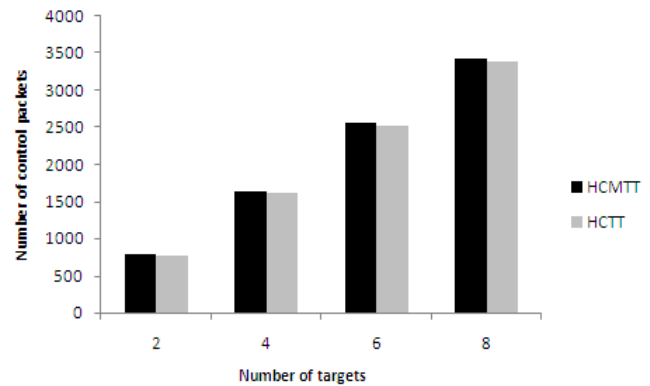


Fig.8. Total number of control packets traversing within the network when the number of targets varies.

- [9] H.-T., Kung, D. Vlah, "Efficient Location Tracking using Sensor Networks," in *Proc. of the IEEE WCNC*, March 2003, pp. 1954-1961.
- [10] F. Lalooses, H. Susanto, and C.-H., Chang, "Recovery Target Tracking in Wildlife", Electrical Engineering Department, Tufts University, Technical Report, 2005.
- [11] S. Park, K. Shin, A. Abraham, S. Han, "Optimized Self Organized Sensor Networks," *Sensors*, Vol 7, pp. 730-742, 2007.
- [12] Y., Xu, J. Winter, W.-C., Lee, "Prediction-based Strategies for Energy Saving in Object Tracking Sensor Networks," in *Proc. of the 2004 IEEE MDM Int. Conf.*, 2004, pp. 346-357.
- [13] Y. Xu, J. Winter, W.-C., Lee, "Dual Prediction-based Reporting for Object Tracking Sensor Networks," in *Proc. of the 1st MobiQuitous Ann. Conf.*, Aug., 2004, pp. 154-163.
- [14] H.-W., Tsai, C.-P., Chu, T.-S., Chen, "Dynamic Object Tracking in Wireless Sensor Networks," *Computer Comm.*, Vol. 30, pp. 1811-1825, 2007.
- [15] L. Yen, B. Wu, C. Yang, "Tree-based object tracking without mobility statistics in wireless sensor networks," *Wirel. Netw.* Vol 16, NO 5, pp. 1263-1276, July 2010.
- [16] H. Yang, B. Sikdar, "A Protocol for Tracking Mobile Targets using Sensor Networks," in *Proc. of the 1st IEEE Int. Workshop on Sensor Network Protocols and Applications*, May, 2003, pp. 71-81.
- [17] I. Lee, Z. Fu, W. Yang, M. Park, "An Efficient Dynamic Clustering Algorithm for Object Tracking in Wireless Sensor Networks," *The 2nd International Conference on Complex Systems and Applications*, June. 8-10, 2007, pp. 1484-1488.
- [18] W.-R., Chang, H.-T., Lin, Z.-Z., Cheng, "CODA: A Continuous Object Detection and Tracking Algorithm for Wireless Ad hoc Sensor Networks," in *Proc. of the 5th IEEE CCNC 2008*, 2008, pp. 168-174.
- [19] G. Jin, X. Lu, M. Park, "Dynamic Clustering for object tracking in wireless sensor networks," in *Proc. of 3rd International Symposium on Ubiquitous computing systems*, Seoul, 2006, pp. 200-209.
- [20] The Castalia Website. [Online]. Available: <http://castalia.npc.nicta.com.au/>
- [21] The OMNet++ Website. [Online]. Available: <http://www.omnetpp.org/>
- [22] "CC2420 data sheet," Texas Instruments, March 2007.
- [23] M. Slavik, I. Mahgoub, A. Badi, M. Ilyas, "Analytical model of energy consumption in hierarchical wireless sensor networks," in *High-Capacity Optical Networks and Enabling Technologies (HONET)*, Dec 2010, pp. 84-90.
- [24] E. Olule, G. Wang, M. Guo and M. Dong, "RARE: An Energy Efficient Target Tracking Protocol for Wireless Sensor Networks," in *International Conference on Parallel Processing Workshops (ICPPW 2007)*, 2007.